Live Mobile Learning System with Resizable Split Panes

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Abstract. Most existing live mobile distance learning systems do not display presentation slides and a whiteboard simultaneously, resulting in a discrepancy from a real classroom where both can be viewed together. Even when both are displayed simultaneously, the limited screen size of mobile devices can significantly compromise their visibility, negatively affecting both instructor's usability and students' comprehension. Therefore, we propose a live mobile distance learning system with a user-adjustable UI that allows instructors to dynamically resize the slides and whiteboard areas. This system allows students to maintain sufficient visibility on limited screen sizes, providing an experience closer to that of a real classroom. The results of our preliminary evaluation show that the proposed system outperforms existing systems in visibility and comprehensibility, demonstrating that a user-adjustable UI can effectively enhance learning performance in mobile remote education environments.

Keywords: Live mobile learning · Whiteboard · Annotation · Resizable split panes

1 Introduction

Distance learning has long been studied as a means of enabling students to attend classes without physically presence. With the widespread use of mobile devices, mobile learning has attracted significant attention from researchers, allowing students to participate in a learning session anytime and anywhere using smartphones or tablets[10].

Early mobile learning systems include Classroom Presenter[6] and MLVLS[9]. Classroom Presenter, a tablet-based live mobile learning system, supports slides sharing with annotation but lacks video and chat features for interaction between a teacher and students. MLVLS, running on Symbian smartphones, did include video, but interaction remains limited, and small screens make it difficult to read detailed slides.

The COVID-19 pandemic dramatically accelerated the adoption of distance learning systems, as distance learning became the only viable alternative to the traditional face-to-face classroom. Since then, live mobile learning has evolved

beyond a temporary backup plan into a new standard in education. During the pandemic, video conferencing platforms such as Cisco Webex[1], Zoom[5], Microsoft Teams[3], Google Meet[2] were widely used for distance learning. These platforms support various features, including slides with annotation, whiteboard, instructor video, chat, and participant management. However, they do not support the simultaneous view of both presentation slides and a whiteboard. Although a simple side-by-side layout is possible, the limited screen size of mobile devices can significantly compromise their visibility. It negatively affects both the instructor's delivery and the students' overall comprehension.

To address this limitation, the Large-Sized Tablet-Based Live Mobile Learning System[7] was proposed. By leveraging the larger display area of tablets, this system enables the simultaneous view of slides and a whiteboard and supports features such as slides with annotation, whiteboard, instructor video, chat, and participant management. in terms of lecture understanding, sophistication of a lecture, and the feedback to a question .

To extend the benefits of the Large-Sized Tablet-Based Live Mobile Learning System, which have been validated on large-sized tablets, to a wider range of general mobile devices, we propose a live mobile learning system with a user-adjustable UI which allows instructors to dynamically resize the slide and whiteboard areas. This flexible layout ensures that complex slides remain sufficiently legible even on compact displays, thereby delivering a learning experience that more closely approximates the effectiveness and interactivity of a traditional classrooms.

2 Live Mobile Learning System with User-Adjustable UI

Figure 1 shows the run-time communication architecture of the proposed system.

The system consists of a server and tablet clients. The server is composed of signaling server and a streaming server. The signaling server is responsible for centralized session management and real-time coordination among clients. It is implemented using the NWProtocolWebSocket of Swift Network Framework[4] and runs as an Xcode Command Line Tool. The signaling server provides multi-classroom functionalities, including classroom creation, termination, joining, leaving and chat. It handles multiple message types by using Codable based enum and manages multiple classrooms using a dictionary keyed by UUIDs. The streaming server operates as a separate process from the signaling server but is integrated into a single Xcode target. It is executed by invoking the NGINX executable along with its configuration file through Swift's Process() API. It uses the RTMP (Real-Time Messaging Protocol)[8] to receive video streams including presentation slides with annotation, whiteboard, instructor camera video, and audio from the instructor client, and delivers them to student clients in real time.

Figure 2 shows the home screen. Users can either join an existing classroom as a student or create a new room using the "+" button to start a lecture as an instructor. The room list is updated in real time via communication with the

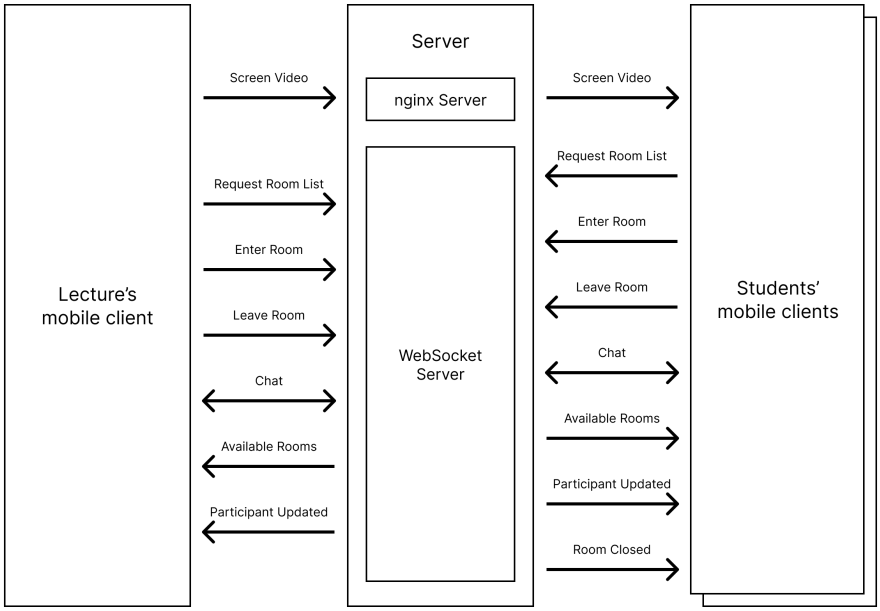

Fig. 1. Communication architecture of the proposed system

signaling server, and users can also manually refresh the list by pulling down the UITableView. Both room creation and entry are handled by sending an Enter Room message to the server.

Figure 3 shows the instructor client's classroom interface. The UI consists of presentation slides, a whiteboard, instructor camera video, a participant list, and a chat. Detailed descriptions of each component are as follows.

Annotation on slides is implemented using PDFPageOverlayViewProvider, where a PKCanvasView of PencilKit[**?**] is added to the `overlayView` of each `PDFPage`. Pen and eraser tools are implemented as toggle buttons, which switch the `PKTool` accordingly. When both tools are disabled, the view behaves as a scrollable PDF document.

3 Preliminary Evaluation

To evaluate the effectiveness of the proposed system in supporting learning and its usability, a user satisfaction study was conducted. The evaluation involved five participants and compared three systems: (1) Cisco Webex, which allows only one of either lecture slides or a whiteboard to be displayed at a time, (2) a system with a fixed 5:5 split UI displaying both slides and a whiteboard simultaneously, and (3) the proposed system with a user-adjustable UI.

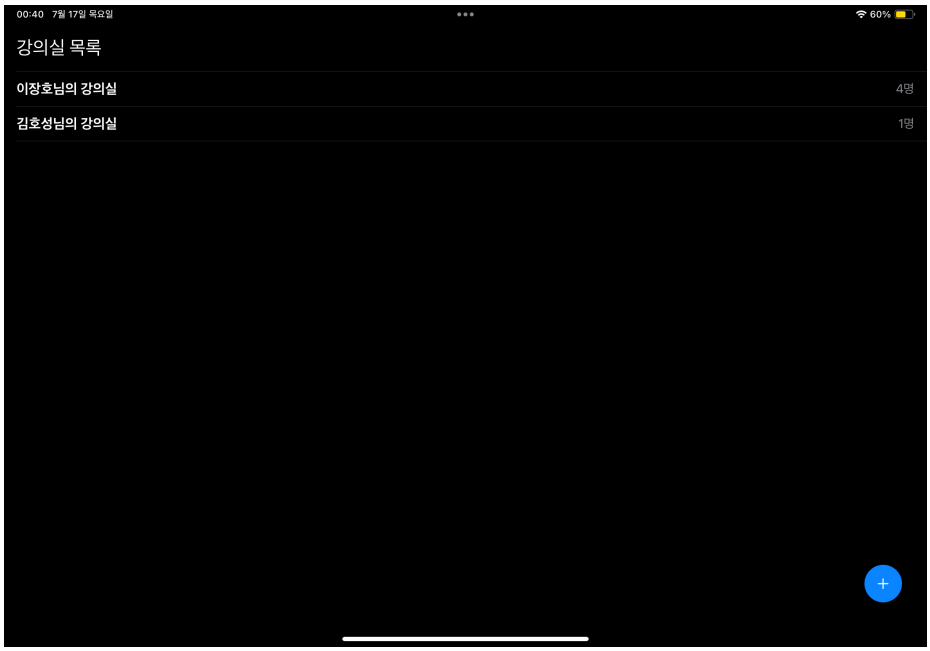

Fig. 2. User interface of a home view

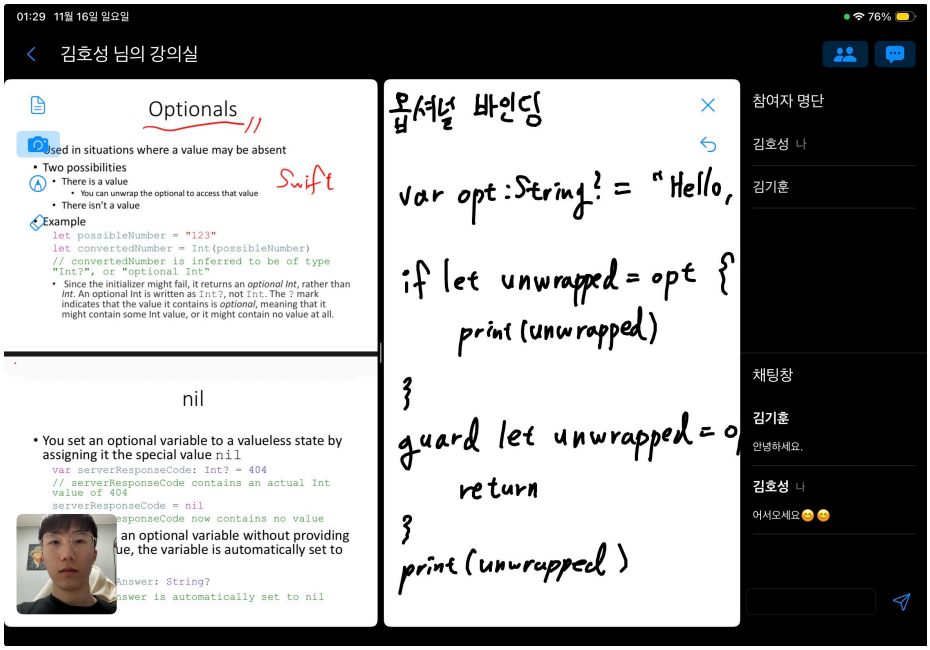

Fig. 3. User interface of a classroom view

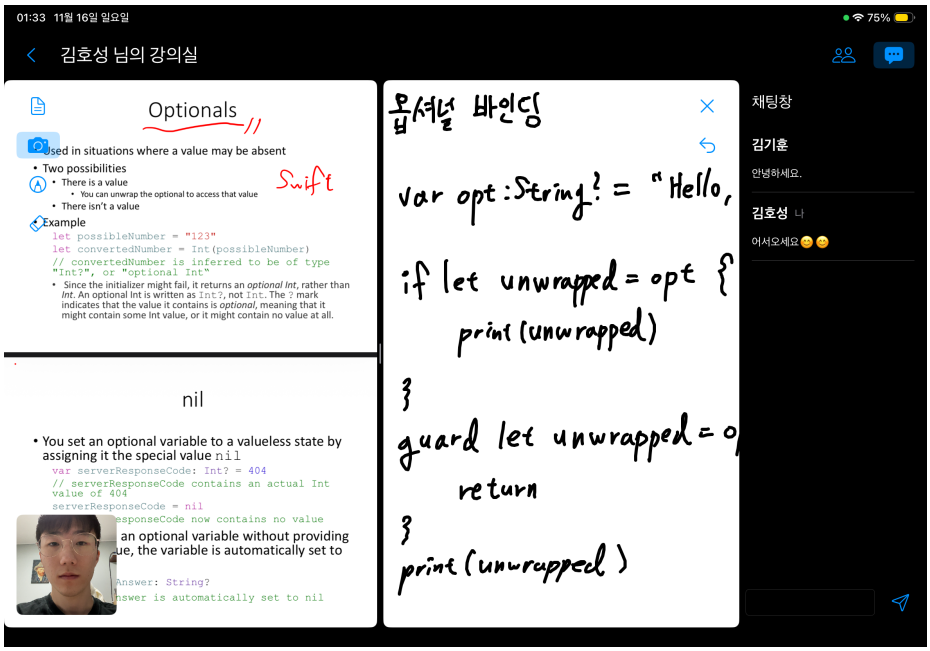

Fig. 4. User interface of a classroom view with participants panel hidden

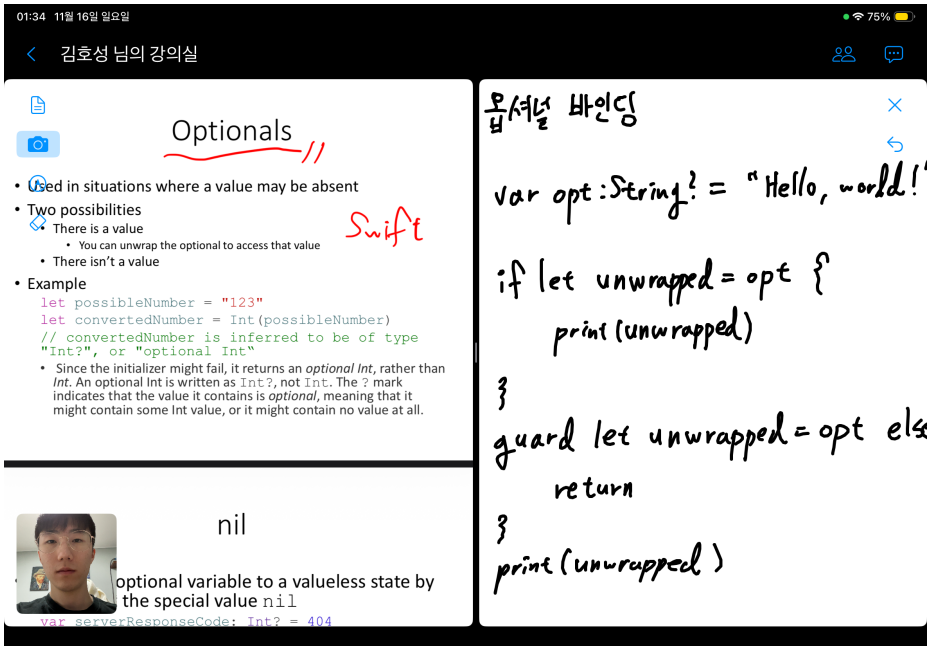

Fig. 5. User interface of a classroom view with participants and chat panels hidden

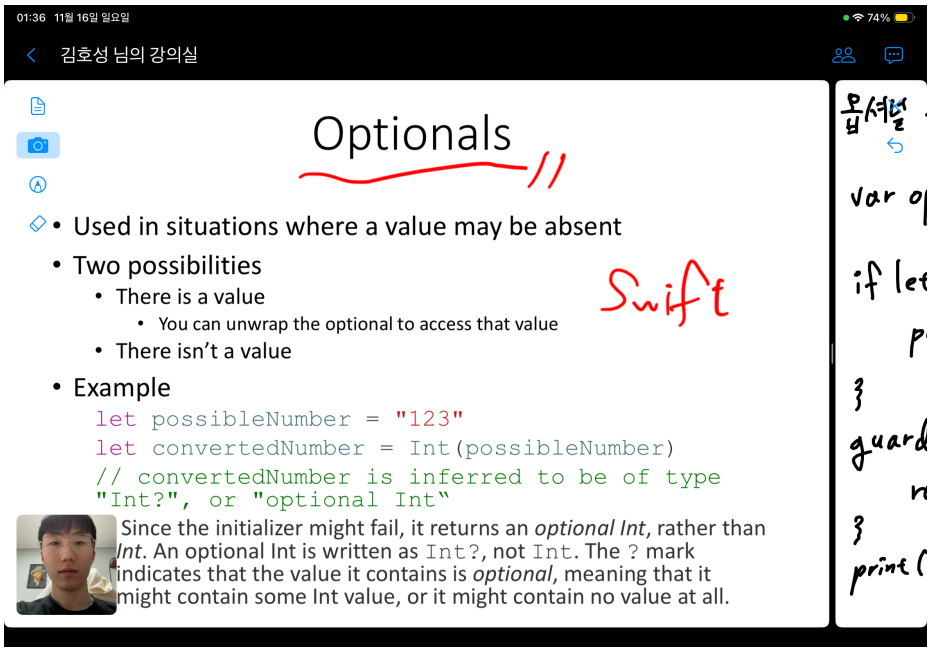

Fig. 6. User interface of a classroom view an expanded presentation slide area

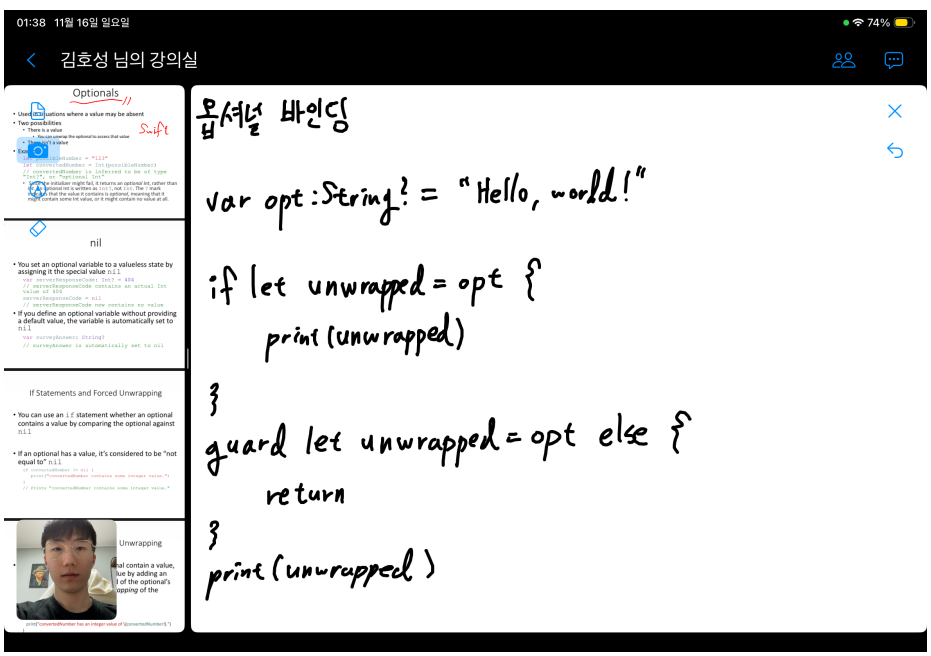

Fig. 7. User interface of a classroom view with an expanded whiteboard area

After conducting lectures using each system, participants were asked to respond to the following three questions: (1) How clearly visible were the lecture slides? (2) How clearly visible was the whiteboard? and (3) How easy was it to understand the lecture when both slides and the whiteboard were used? Each question was rated on a 5-point Likert scale, where 1 indicates very poor and 5 indicates excellent. The results are summarized in Tables 1, 2, and 3.

Tables 1 and 2 present the results for slide visibility and whiteboard visibility, respectively. Webex received consistently high scores in both categories, indicating that displaying only one content element at a time ensures sufficient readability. In contrast, the fixed 5:5 UI system received significantly lower scores, as both the slide and whiteboard areas were constrained to smaller sizes in order to be displayed simultaneously.

The proposed user-adjustable UI system achieved high scores in both categories, comparable to Webex. This suggests that allowing instructors to dynamically adjust the size of the slide and whiteboard areas effectively maintains readability for both types of content, depending on the lecture context.

Table 1. Respondents' perception of the visibility of a lecture slide

Target System	Respondents					Mean	SD
	A	B	C	D	E		
Webex	5	5	5	5	5	5	0
5:5 Fixed UI	1	5	1	2	4	2.6	1.82
User-Adjustable UI	5	5	5	5	5	5	0

Table 2. Respondents' perception of the visibility of a whiteboard

Target System	Respondents					Mean	SD
	A	B	C	D	E		
Webex	3	5	5	4	4	4.2	0.84
5:5 Fixed UI	1	3	2	3	5	2.8	1.49
User-Adjustable UI	5	4	4	5	5	4.6	0.55

Table 3 presents the results for lecture comprehensibility when both slides and the whiteboard are used together. Webex received a relatively high average score of 4.0; however, some participants reported inconvenience due to the inability to use slides and the whiteboard simultaneously, resulting in a slight reduction in score. The fixed 5:5 UI system received the lowest average score of 2.4, with participants citing reduced readability of slides as a major factor negatively affecting their learning experience.

In contrast, the proposed system achieved the highest average score of 4.6. Participants noted that the ability to dynamically adjust the relative sizes of the slide and whiteboard allowed them to focus on the most relevant content at any given moment. For example, slides could be enlarged during explanation-heavy segments, while the whiteboard could be expanded during problem-solving or annotation phases. This flexibility contributed to improved overall lecture comprehension.

Table 3. Respondents' perception of the comprehensibility of a lecture with a lecture slide and a whiteboard combined

Target System	Respondents					Mean	SD
	A	B	C	D	E		
Webex	5	5	3	4	3	4	1
5:5 Fixed UI	3	2	2	1	4	2.4	1.15
User-Adjustable UI	4	4	5	5	5	4.6	0.55

Overall, the results demonstrate that the proposed user-adjustable UI system provides higher user satisfaction compared to both the conventional remote learning system (Webex) and the fixed 5:5 UI system. While Webex ensures high readability by displaying a single content type, it lacks the ability to present slides and whiteboard content simultaneously. On the other hand, the fixed 5:5 UI system enables simultaneous display but suffers from reduced readability due to limited screen space.

The proposed system effectively addresses these limitations by enabling both simultaneous presentation and flexible resizing of content. As a result, it provides an effective balance between visibility and functionality, leading to an improved learning experience on mobile devices.

4 Conclusion

In this paper, we proposed and implemented a mobile distance learning system with a user-adjustable UI to address the limitation of existing systems that fail to effectively present both lecture slides and a whiteboard simultaneously. The proposed system allows instructors to dynamically control the relative sizes of these two content areas through a grabber interface, enabling learners to view essential information at an appropriate scale even on limited mobile screen sizes.

In addition, collapsible participant and chat panels were designed to minimize unnecessary screen occupancy, thereby improving the utilization of limited display space and enhancing visual focus on core learning content.

The results of the user satisfaction study demonstrate that the proposed system outperforms existing approaches in terms of slide visibility, whiteboard visibility, and overall lecture comprehension. These findings indicate that the

user-adjustable UI effectively improves the learning experience in mobile distance learning environments.

The full source code of the proposed system is publicly available at: https://github.com/H0sungKim/CollaborativeComputingLab

Acknowledgments. Acknowledgements

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